TOYWORLD – E-COMMERCE MANAGEMENT SYSTEM

DATABASE RELATIONSHIP ANALYSIS USING JOINS

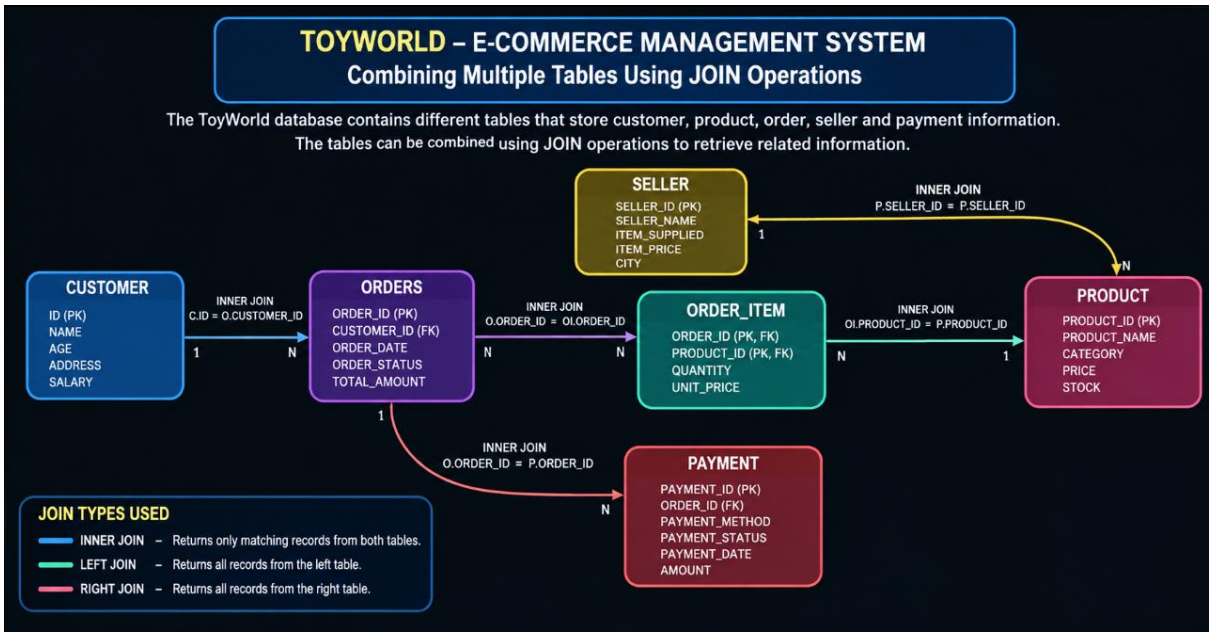
Objectives

- To manage customer details in the ToyWorld E-Commerce System.
- To store and maintain product information.
- To manage customer orders and order details.
- To maintain seller and payment information.
- To retrieve customer purchase details using SQL queries.
- To generate useful business reports.

1. Combining Multiple Tables

The toyworld database contains different tables that store customer, product, order, seller, and payment information.

The tables can be combined using join operations to retrieve related information.



2. Inner Join

Inner join returns only the records that have matching values in both tables.

2.1 Customer and Orders

```
SQL> SELECT
  2      C.ID,
  3      C.NAME,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      O.ORDER_STATUS,
  7      O.TOTAL_AMOUNT
  8  FROM CUSTOMER C
  9  INNER JOIN ORDERS O
 10  ON C.ID = O.CUSTOMER_ID;
```

ID	NAME	ORDER_ID	ORDER_DAT	ORDER_STATUS	TOTAL_AMOUNT
1	Ramesh	1001	20-SEP-26	Confirmed	1350
2	Khilan	1002	28-SEP-26	Shipped	850
3	kaushik	1003	22-SEP-26	Delivered	1500
4	Chaitali	1004	23-SEP-26	Pending	650
5	Hardik	1005	24-SEP-26	Confirmed	950
1	Ramesh	1006	25-SEP-26	Shipped	1100
2	Khilan	1007	26-SEP-26	Delivered	1800
3	kaushik	1008	27-SEP-26	Confirmed	750

8 rows selected.

This displays customers who have placed orders.

2.2 Orders and Payment

```
SQL> SELECT
  2      O.ORDER_ID,
  3      O.CUSTOMER_ID,
  4      O.ORDER_DATE,
  5      O.TOTAL_AMOUNT,
  6      P.PAYMENT_ID,
  7      P.PAYMENT_METHOD,
  8      P.PAYMENT_STATUS
  9  FROM ORDERS O
 10  INNER JOIN PAYMENT P
 11  ON O.ORDER_ID = P.ORDER_ID;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	PAYMENT_ID	PAYMENT_METHOD	PAYMENT_STATUS
1001	1	20-SEP-26	1350	1	UPI	Successful
1002	2	28-SEP-26	850	2	Credit Card	Successful
1003	3	22-SEP-26	1500	3	Debit Card	Failed
1004	4	23-SEP-26	650	4	Cash on Delivery	Successful
1005	5	24-SEP-26	950	5	UPI	Successful
1006	1	25-SEP-26	1100	6	Credit Card	Failed
1007	2	26-SEP-26	1800	7	Debit Card	Successful
1008	3	27-SEP-26	750	8	UPI	Successful

8 rows selected.

This displays order details and their corresponding payment information.

2.3 Product and Order Details

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      OI.ORDER_ID,
  5      OI.QUANTITY,
  6      OI.UNIT_PRICE
  7  FROM PRODUCT P
  8  INNER JOIN ORDER_ITEM OI
  9  ON P.PRODUCT_ID = OI.PRODUCT_ID;
```

```
PRODUCT_ID PRODUCT_NAME                                                                                          ORDER_ID   QUANTITY UNIT_PRICE
---------- ---------------------------------------------------------------------------------------------------- ---------- ---------- ----------
       101 ABC Learning Kit                                                                                           1001          2        300
       103 Remote Car                                                                                                 1002          4        850
       104 Chess Board                                                                                                1003          1       1500
       105 Football                                                                                                   1004          1        650
       109 Superhero Figure                                                                                           1006          2        550
       101 ABC Learning Kit                                                                                           1007          3        600
       103 Remote Car                                                                                                 1008          1        750

7 rows selected.
```

This displays products that have been included in customer orders.

3. Left Join

Left join returns all records from the left table and matching records from the right table.

3.1 Display all Customers and their Orders

```
SQL> SELECT
  2      C.ID,
  3      C.NAME,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      O.ORDER_STATUS,
  7      O.TOTAL_AMOUNT
  8  FROM CUSTOMER C
  9  LEFT JOIN ORDERS O
 10  ON C.ID = O.CUSTOMER_ID;
```

```
        ID NAME                           ORDER_ID ORDER_DAT ORDER_STATUS         TOTAL_AMOUNT
---------- ------------------------------ ---------- --------- -------------------- ------------
         1 Ramesh                               1001 20-SEP-26 Confirmed                    1350
         2 Khilan                               1002 28-SEP-26 Shipped                       850
         3 kaushik                              1003 22-SEP-26 Delivered                    1500
         4 Chaitali                             1004 23-SEP-26 Pending                       650
         5 Hardik                               1005 24-SEP-26 Confirmed                     950
         1 Ramesh                               1006 25-SEP-26 Shipped                      1100
         2 Khilan                               1007 26-SEP-26 Delivered                    1800
         3 kaushik                              1008 27-SEP-26 Confirmed                     750

8 rows selected.
```

This displays all customers, including customers who have not placed any orders.

4. Right Join

Right join returns all records from the right table and matching records from the left table.

4.1 Display all Orders and Customer Details

```
SQL> SELECT
  2       C.ID,
  3       C.NAME,
  4       O.ORDER_ID,
  5       O.ORDER_DATE,
  6       O.ORDER_STATUS,
  7       O.TOTAL_AMOUNT
  8  FROM CUSTOMER C
  9  RIGHT JOIN ORDERS O
 10  ON C.ID = O.CUSTOMER_ID;
```

```
        ID NAME                           ORDER_ID ORDER_DAT ORDER_STATUS         TOTAL_AMOUNT
---------- ------------------------------ ---------- --------- -------------------- ------------
         1 Ramesh                           1001 20-SEP-26 Confirmed                    1350
         1 Ramesh                           1006 25-SEP-26 Shipped                      1100
         2 Khilan                           1002 28-SEP-26 Shipped                       850
         2 Khilan                           1007 26-SEP-26 Delivered                    1800
         3 kaushik                          1003 22-SEP-26 Delivered                    1500
         3 kaushik                          1008 27-SEP-26 Confirmed                     750
         4 Chaitali                         1004 23-SEP-26 Pending                       650
         5 Hardik                           1005 24-SEP-26 Confirmed                     950

8 rows selected.
```

This displays all orders along with their corresponding customer details.

5. Complete Order Details

To display complete order information by combining Customer, Orders, Order_Item and Product tables.

```
SQL> SELECT
  2  C.ID,
  3  C.NAME,
  4  O.ORDER_ID,
  5  O.ORDER_DATE,
  6  O.ORDER_STATUS,
  7  P.PRODUCT_NAME,
  8  OI.QUANTITY,
  9  OI.UNIT_PRICE,
 10  O.TOTAL_AMOUNT
 11  FROM CUSTOMER C
 12  JOIN ORDERS O
 13  ON C.ID = O.CUSTOMER_ID
 14  JOIN ORDER_ITEM OI
 15  ON O.ORDER_ID = OI.ORDER_ID
 16  JOIN PRODUCT P
 17  ON OI.PRODUCT_ID = P.PRODUCT_ID;
```

```
        ID NAME                           ORDER_ID ORDER_DAT ORDER_STATUS         PRODUCT_NAME
  QUANTITY UNIT_PRICE TOTAL_AMOUNT
---------- ------------------------------ ---------- --------- -------------------- ----------------------
-------------------------- ---------- ----------
         1 Ramesh                           1001 20-SEP-26 Confirmed            ABC Learning Kit
 2         300         1350
         2 Khilan                           1007 26-SEP-26 Delivered            ABC Learning Kit
 3         600         1800
         2 Khilan                           1002 28-SEP-26 Shipped              Remote Car
 4         850          850
         3 kaushik                          1008 27-SEP-26 Confirmed            Remote Car
 1         750          750
         3 kaushik                          1003 22-SEP-26 Delivered            Chess Board
 1        1500         1500
         4 Chaitali                         1004 23-SEP-26 Pending              Football
 1         650          650
         1 Ramesh                           1006 25-SEP-26 Shipped              Superhero Figure
 2         550         1100

7 rows selected.
```

This displays complete customer order details, including product names, quantity and price.

6. Customer Purchase History

To display the total purchase amount of each customer.

```
SQL> SELECT
  2      C.ID,
  3      C.NAME,
  4      COUNT(O.ORDER_ID) AS TOTAL_ORDERS,
  5      SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  6  FROM CUSTOMER C
  7  JOIN ORDERS O
  8  ON C.ID = O.CUSTOMER_ID
  9  GROUP BY C.ID, C.NAME;
```

```
        ID NAME                           TOTAL_ORDERS TOTAL_PURCHASE
---------- ------------------------------ ------------ --------------
         1 Ramesh                                    2           2450
         2 Khilan                                    2           2650
         3 kaushik                                   2           2250
         4 Chaitali                                  1            650
         5 Hardik                                    1            950
```

This displays the number of orders and total purchase amount of each customer.

7. Customer Purchase History with Payment

```
SQL> SELECT
  2      C.ID,
  3      C.NAME,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      P.PRODUCT_NAME,
  7      OI.QUANTITY,
  8      OI.UNIT_PRICE,
  9      O.TOTAL_AMOUNT,
 10      PY.*
 11  FROM CUSTOMER C
 12  INNER JOIN ORDERS O
 13      ON C.ID = O.CUSTOMER_ID
 14  INNER JOIN ORDER_ITEM OI
 15      ON O.ORDER_ID = OI.ORDER_ID
 16  INNER JOIN PRODUCT P
 17      ON OI.PRODUCT_ID = P.PRODUCT_ID
 18  INNER JOIN PAYMENT PY
 19      ON O.ORDER_ID = PY.ORDER_ID
 20  ORDER BY C.ID;
```

```
       ID NAME                           ORDER_ID ORDER_DAT PRODUCT_NAME
    QUANTITY UNIT_PRICE TOTAL_AMOUNT PAYMENT_ID   ORDER_ID
---------- ---------- ------------ ---------- ---------- ----------------------------------------------------
------- ---------- ---------- ------------ ---------- ----------
PAYMENT_METHOD                                PAYMENT_STATUS                 PAYMENT_D     AMOUNT
--------------------------------------------- ------------------------------ --------- ----------
         1 Ramesh                           1001 20-SEP-26 ABC Learning Kit
    2         300         1350          1       1001
UPI                                           Successful                     01-SEP-26        599

         1 Ramesh                           1006 25-SEP-26 Superhero Figure
    2         550         1100          6       1006
Credit Card                                   Failed                         06-SEP-26        650

         2 Khilan                           1002 28-SEP-26 Remote Car
    4         850          850          2       1002
Credit Card                                   Successful                     02-SEP-26        899

         2 Khilan                           1007 26-SEP-26 ABC Learning Kit
    3         600         1800          7       1007
Debit Card                                    Successful                     07-SEP-26        999

         3 kaushik                          1003 22-SEP-26 Chess Board
    1        1500         1500          3       1003
Debit Card                                    Failed                         03-SEP-26        450

         3 kaushik                          1008 27-SEP-26 Remote Car
    1         750          750          8       1008
UPI                                           Successful                     08-SEP-26        550

         4 Chaitali                         1004 23-SEP-26 Football
    1         650          650          4       1004
Cash on Delivery                              Successful                     04-SEP-26        750


7 rows selected.
```

8. Multi-Table Business Report

To combine multiple tables and generate a business report.

```
SQL> SELECT
  2      C.NAME,
  3      O.ORDER_ID,
  4      O.ORDER_DATE,
  5      P.PRODUCT_NAME,
  6      OI.QUANTITY,
  7      OI.UNIT_PRICE,
  8      O.TOTAL_AMOUNT
  9  FROM CUSTOMER C
 10  JOIN ORDERS O
 11  ON C.ID = O.CUSTOMER_ID
 12  JOIN ORDER_ITEM OI
 13  ON O.ORDER_ID = OI.ORDER_ID
 14  JOIN PRODUCT P
 15  ON OI.PRODUCT_ID = P.PRODUCT_ID
 16  ORDER BY O.ORDER_ID;
```

```
NAME                           ORDER_ID ORDER_DAT PRODUCT_NAME
UANTITY UNIT_PRICE TOTAL_AMOUNT
------------------------------ ---------- --------- ----------------------
------- ---------- ------------
Ramesh                               1001 20-SEP-26 ABC Learning Kit
 2         300         1350
Khilan                               1002 28-SEP-26 Remote Car
 4         850          850
kaushik                              1003 22-SEP-26 Chess Board
 1        1500         1500
Chaitali                             1004 23-SEP-26 Football
 1         650          650
Ramesh                               1006 25-SEP-26 Superhero Figure
 2         550         1100
Khilan                               1007 26-SEP-26 ABC Learning Kit
 3         600         1800
kaushik                              1008 27-SEP-26 Remote Car
 1         750          750

7 rows selected.
```

This displays a business report containing customer, order and product information.

9. Payment Transaction Report

To display payment transaction details.

```
SQL> SELECT
  2      PY.*,
  3      O.CUSTOMER_ID,
  4      O.ORDER_DATE,
  5      O.ORDER_STATUS,
  6      O.TOTAL_AMOUNT
  7  FROM PAYMENT PY
  8  INNER JOIN ORDERS O
  9  ON PY.ORDER_ID = O.ORDER_ID
 10  ORDER BY PY.ORDER_ID;
```

```
PAYMENT_ID   ORDER_ID PAYMENT_METHOD                                PAYMENT_STATUS               PAYMENT_D     AMOUNT CUSTOMER_ID ORDER_DAT ORDER_STA
TUS     TOTAL_AMOUNT
---------- ---------- -------------------------------------------- ---------- ---------- ---------- ---------- ---------
----------- ------------
         1       1001 UPI                                          Successful                01-SEP-26        599           1 20-SEP-26 Confirmed
1350
         2       1002 Credit Card                                  Successful                02-SEP-26        899           2 28-SEP-26 Shipped
 850
         3       1003 Debit Card                                   Failed                    03-SEP-26        450           3 22-SEP-26 Delivered
1500
         4       1004 Cash on Delivery                             Successful                04-SEP-26        750           4 23-SEP-26 Pending
 650
         5       1005 UPI                                          Successful                05-SEP-26       1200           5 24-SEP-26 Confirmed
 950
         6       1006 Credit Card                                  Failed                    06-SEP-26        650           1 25-SEP-26 Shipped 1
 100
         7       1007 Debit Card                                   Successful                07-SEP-26        999           2 26-SEP-26 Delivered
1800
         8       1008 UPI                                          Successful                08-SEP-26        550           3 27-SEP-26 Confirmed
 750

8 rows selected.
```

Conclusion

The ToyWorld E-Commerce Management System helps manage customer, product, seller, order, and payment information efficiently. SQL queries and JOIN operations help retrieve

related information from multiple tables and generate useful reports. This project demonstrates how a database can be used to manage an e-commerce business effectively.